HelixGitpx — Documentation Suite

A self-healing, AI-assisted, multi-upstream Git federation platform.

Part of the Helix Development ecosystem.

Document Classification

Attribute	Value
Document Class	Enterprise Engineering Specification Suite
Project Codename	`helixgitpx`
Parent Program	Helix Development Platform
Version	v1.0.0 (Implementation-Ready)
Status	`APPROVED` — supersedes `Git_Proxy_Master_Specification.md`
Owner	vasic-digital
License	Apache-2.0 (code) / CC-BY-SA-4.0 (docs)

How to Read This Suite

HelixGitpx is a large, multi-service, multi-platform system. The documentation is structured so that **each role can find everything they need without reading everything**. Use this index as your routing table.

Role	Start Here	Then Read
Executive / Product	01-vision-scope-constraints.md	13-roadmap/17-milestones.md
Architect	02-system-architecture.md	03-microservices-catalog.md, 14-diagrams/
Backend Engineer	03-microservices-catalog.md	04-data-model.md, 05-event-streaming.md, 06-grpc-api.md

Role	Start Here	Then Read
Frontend Engineer (Web)	12-frontend-angular.md	07-rest-api.md, 08-live-events.md
Mobile / Desktop Engineer	13-mobile-desktop-kmp.md	06-grpc-api.md
AI / ML Engineer	10-llm-self-learning.md	09-conflict-resolution.md
Security Engineer	11-security-compliance.md	15-testing-strategy.md
SRE / DevOps	16-infrastructure-scaling.md	17-devops-cicd.md, 18-observability.md, 19-operations-runbook.md
QA Engineer	15-testing-strategy.md	All API docs
Contributor / OSS Dev	20-developer-guide.md	02-system-architecture.md

Complete Document Index

00 — Core

#	Document	Purpose
01	Vision, Scope & Constraints	Product mandate, non-negotiable requirements, success metrics, out-of-scope
02	Glossary	Terminology reference

01 — Architecture

#	Document	Purpose
02	System Architecture	C4 L1-L3, bounded contexts, cross-cutting concerns
—	ADR Log	Architecture Decision Records

02 — Services

#	Document	Purpose
03	Microservices Catalog	All 18 services with responsibilities, interfaces, scaling profile
09	Conflict Resolution Engine	Multi-way merge, CRDT metadata, three-phase reconciliation
10	Git Provider Integrations	Adapter matrix for GitHub / GitLab / Gitee / GitFlic / GitVerse / BitBucket / …
23	WASM Plugin SDK	Third-party-extensible adapters, normalisers, validators

03 — Data & Events

#	Document	Purpose
04	Data Model (PostgreSQL)	Full schema, indexes, partitions, RLS policies
05	Event Streaming (Kafka)	Topics, schemas, consumer groups, retention, compaction
05b	Search & Indexing (OpenSearch/Meilisearch/Qdrant)	Index strategy, synonyms, vector search

04 — APIs

#	Document	Purpose
06	gRPC API Specification	Protobuf schemas, services, streaming
07	REST API Specification	OpenAPI 3.1, auth, pagination, idempotency
08	Live Events & WebSocket Fallback	Subscription model, resume tokens, backpressure
29	API Versioning & Deprecation	Support windows, breaking change policy, rollout mechanics

05 — Frontend (Web)

#	Document	Purpose
12	Angular Web Application	Architecture, state, design system, a11y, PWA
28	Accessibility Standards (WCAG 2.2 AA)	Design, testing, manual/automated audits, feedback intake
32	Internationalisation & Localisation	20 locales at GA, ICU MessageFormat, RTL, Crowdin process
37	Design System & Tokens	Colour / typography / spacing tokens, cross-platform pipeline
37	Design System & Tokens	Colour, typography, spacing, motion; cross-platform token pipeline

06 — Mobile & Desktop

#	Document	Purpose
13	Mobile & Desktop (Kotlin Multiplatform + Compose)	Shared UI + logic across Android / iOS / Windows / macOS / Linux

07 — AI / ML

#	Document	Purpose

#	Document	Purpose
10	Self-Learning LLM Platform	Continuous feedback, RLAIF, model registry, drift detection

08 — Security

#	Document	Purpose
11	Security & Compliance	SonarQube, Snyk, zero-trust, SPIFFE/SPIRE, SLSA L3, SOC 2 readiness
24	Threat Model	STRIDE per trust boundary, MITRE ATT&CK mapping, residual risk
27	Data Retention, Privacy & DSR	GDPR / CCPA, retention matrix, data-subject rights workflows
36	Trust Center	Public security + privacy posture, certifications, disclosures

09 — Observability

#	Document	Purpose
18	Observability Platform	OTel end-to-end, SLOs, SLIs, dashboards, alerts, runbooks

10 — Testing

#	Document	Purpose
15	Testing Strategy (100% Coverage)	Unit / integration / E2E / fuzz / property / security / DDoS / chaos / perf / DR
33	Performance Baselines & Benchmarks	Targets, measured baselines, regression guardrails

11 — DevOps

#	Document	Purpose
16	Infrastructure & Scaling	K8s topology, multi-region, auto-scaling, cost optimization
17	DevOps & CI/CD	GitOps (Argo CD), pipelines, release trains, feature flags
26	On-Prem & Air-Gapped Deployment	Customer-managed installations, signed bundles, FIPS mode
31	Release Management & Branching	Trunk-based, release trains, canaries, hotfix flow, rollback
34	Cost Optimisation	FinOps principles, tenant attribution, levers with impact

12 — Operations

#	Document	Purpose

#	Document	Purpose
19	Operations Runbook	Incident response, on-call rotation, maintenance windows
20	Developer Guide	Local dev setup, contribution workflow, code style
21	User Guide	End-user documentation with screenshots
22	Billing & Metering	Plans, quotas, usage events, invoicing, dunning
25	Migration Guide	Onto (and off) HelixGitpx from GitHub / GitLab / Bitbucket / etc.
30	Service Level Agreements	Per-plan uptime, support response, credits, compliance
35	Support Handbook	Intake, triage, canned responses, playbooks, escalation
—	runbooks/	Standalone pages: RB-010, RB-011, RB-021, RB-100, RB-120, RB-130, RB-140, RB-150, RB-200, RB-301

13 — Roadmap

#	Document	Purpose
17	Milestones, Phases & Tasks	8 milestones, 32 phases, 500+ tasks, dependency graph
—	RFCs	Open RFCs for future capabilities

14 — Diagrams

Mermaid source for every diagram referenced in the docs. See 14-diagrams/README.md.

15 — Reference

Content	Purpose
ADR Index	All 26 ADRs with context, decisions, and alternatives
ADR Template	Standalone template with writing tips
Glossary	Canonical terms
Error Catalog	Stable error codes by domain
Metrics Catalog	Full metrics inventory with labels
Alerts Catalog	Every alert with expression and runbook link
Kafka Topic Catalog	Every topic: producers, consumers, keys, retention, ACLs
Scaffold Catalog	Service / proto / event / UI scaffolds + official integrations
ADR Template	Standalone ADR template for future decisions

Content	Purpose
SDK Examples	Go, TypeScript, Kotlin, Swift, Python example apps
Feature Flag Catalog	Every flag with owner, strategy, sunset date
Compliance Controls	SOC 2 / ISO 27001 / NIST CSF / GDPR mapping
Chaos Playbook	Chaos experiments by tier + Game Days
Policies: Rego	OPA authz + conflict-resolution policies
Policies: Kyverno	K8s admission policies (signing, PSS, hardening)
Load Tests: k6	API mix + git-push fan-out scripts with SLO thresholds

16-18 — Machine-Readable Artefacts

Path	Content
16-schemas/001_auth.sql	Auth context DDL (users, sessions, MFA, PATs, login attempts)
16-schemas/002_repo.sql	Org / repo / upstream / binding DDL with RLS
16-schemas/003_conflict_collab.sql	Conflict, PR, issue, CRDT, AI DDL
16-schemas/004_audit_billing_notify_policy.sql	Audit (append-only), billing, notify, policy, sync DDL
16-schemas/events-avro.json	Avro subjects for Kafka topics
17-protos/common.proto	Shared types (UUID, Page, Actor, Error, enums, Envelope)
17-protos/auth.proto	AuthService (login, refresh, MFA, PAT, sessions, OIDC)
17-protos/repo.proto	RepoService (CRUD, refs, protection, tree/blob, commits, WatchRepo)
17-protos/upstream.proto	UpstreamService + BindingService + provider capabilities
17-protos/sync.proto	SyncService (triggers, jobs, DLQ)
17-protos/conflict.proto	ConflictService (propose, apply, undo, escalate, feedback)
17-protos/events.proto	EventsService (bidi stream, replay)
17-protos/platform.proto	PolicyService + AuditService + NotifyService + BillingService
17-protos/collab.proto	PRService, IssueService, AIService
18-manifests/docker-compose.yml	Local dev stack
18-manifests/Chart.yaml	Helm umbrella chart with dependencies

Path	Content
18-manifests/values-staging.yaml	Staging environment values
18-manifests/values-prod-eu.yaml	Production eu-west-1 values with FIPS, residency pinning
18-manifests/deployment-sample.yaml	Production Deployment + PDB + ConfigMap
18-manifests/hpa-keda-samples.yaml	Autoscaling: HPA + KEDA ScaledObject + ScaledJob
18-manifests/network-policy-samples.yaml	Zero-trust K8s + Cilium policies
18-manifests/argo-application.yaml	GitOps ApplicationSet + canary analysis
18-manifests/kustomize-overlay.yaml	Environment overlay patches on top of Helm-rendered base
18-manifests/terraform/	AWS provisioning module (VPC + EKS + RDS/MSK + S3 + KMS + IRSA)
18-manifests/service-template/	Dockerfile + Makefile + .golangci.yml for new Go services
18-manifests/observability/	Prometheus recording + alerting rules + Grafana RED dashboard
18-manifests/examples/	Temporal FanOutPush workflow + Debezium connector configs
18-manifests/service-template/	Canonical service Dockerfile, Makefile, golangci-lint config
18-manifests/observability/	Prometheus recording + alerting rules, Grafana dashboard JSON
18-manifests/examples/	Temporal workflow (FanOutPush) + Debezium connector configs

Root

Document	Purpose
CONTRIBUTING.md	How to contribute (issues, PRs, CLA / DCO, review process)
CODE_OF_CONDUCT.md	Contributor Covenant v2.1 adapted + enforcement ladder

Quick Facts

Item	Value
Language (backend)	Go 1.23+
HTTP framework	Gin Gonic
RPC	gRPC + gRPC-Web

Item	Value
Async messaging	Apache Kafka + Schema Registry
Primary datastore	PostgreSQL 16 + Timescale extension
Cache / coordination	Redis 7 / Dragonfly
Search	OpenSearch (logs, full-text), Meilisearch (UI), Qdrant (vector)
AI	Self-learning fine-tuned LLMs + LiteLLM router + RLAIF
Web	Angular 19 + NgRx + Tailwind + Nx
Mobile / Desktop	Kotlin Multiplatform + Compose Multiplatform (Android / iOS / Windows / macOS / Linux)
Container runtime	Kubernetes 1.31 + Istio + cert-manager + Argo CD
Code quality	SonarQube, Snyk, Semgrep, Trivy, Gitleaks
Observability	OpenTelemetry → Prometheus + Tempo + Loki + Pyroscope
Supply chain	SLSA L3, Cosign keyless, in-toto, SBOM (CycloneDX)
Test coverage target	≥ 100% (per §15)
Microservices count	18 core + 7 platform
Supported Git providers	12 at GA, extensible via WASM plugins
Scale target	100 k orgs / 10 M repos / 1 B events/day

Versioning

All documents use **semver**. A **MAJOR** bump means breaking changes to public APIs or on-disk formats. A **MINOR** bump means additive changes. A **PATCH** is editorial.

Artifact	Current Version
Documentation Suite	v1.0.0
Protobuf APIs	v1
REST API	v1
PostgreSQL schema	v1.0.0
Kafka event schemas	v1
Web client	v1.0.0
KMP client	v1.0.0

How We Verified the Plan

Every recommendation in this suite has been cross-checked against:

1. The prior **Git Proxy Master Specification v4.0.0** (which this supersedes).
2. Known-good production patterns from large-scale Git platforms.
3. Current (2026) state of the OSS ecosystem — every tool named is actively maintained.
4. The explicit technical constraints set by the project owner.
5. Enterprise requirements (SOC 2, ISO 27001, SLSA, SBOM, a11y WCAG 2.2 AA).

Any section marked **[VERIFY-AT-INTEGRATION]** is a place where the underlying facts change frequently (pricing, API versions, cloud quotas) and must be re-checked when that work is actually scheduled.

— End of README —